



Whirling maple seeds create vortex to fly high and far

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People have always been fascinated by the whirling talent of maple tree seeds. Now a team of Dutch and US researchers sheds light on this truly aerodynamic enigma: how the seed's whirling, called auto-rotation, gives it extra lift allowing it to 'fly' greater distances and drop to the ground slowly. The results, published in the journal *Science*, show how

evolutionary development has enabled plants and animals to improve their flight performance.

The researchers from Wageningen University in the Netherlands and the California Institute of Technology (Caltech) in the US discovered that the whirling action of the maple seeds produces a leading edge vortex - a spinning horizontal tunnel of air along the wing.

Air pressure is lowered over the upper surface of the maple seed, which draws the wind upward to oppose gravity and give it a boost, according to the researchers. The vortex doubles the lift produced by the seeds, against seeds that do not swirl. The vortex's ability to magnify the lift can be compared to the mechanism used by birds, insects and bats when they flap their wings to help them hover.

The researchers calculated the flow of air generated by the swirling seeds by constructing plastic models of the seeds with radiuses of around 12.7 centimetres (i.e. 5 to 10 times larger than a maple seed). The seeds were spun through a large tank of mineral oil using a robot designed at Caltech.

The study showed that the characteristics of the fluid flow produced by the seed

models were the same as those generated by real maple seeds. The only difference was that the flow was through oil, not air.

The researchers then used a powerful laser to produce a sheet of light that illuminated tiny glass beads added to the oil. A camera allowed them to obtain images of the beads' motion during the seed's spinning in the tank. The images showed the presence of a vortex lying near the front leading edge of the spinning seed.

The Dutch researchers investigated the flow created by real maple seeds as they spin freely in order to confirm the results obtained with the robot seed models. The Wageningen researchers constructed a wind tunnel to examine the flow created by real maple seeds, and used smoke to visualise the flow of air around the spinning seeds.

'Maple seeds could represent the most basic and simple design for a miniature helicopter, if the swirling wing could be powered by a micromotor,' explained project co-leader Professor David Lentink of Wageningen University.

Project co-leader Michael H. Dickinson, Professor of Bioengineering at Caltech, said: 'There is enormous interest in the development of micro air vehicles, which, because of their size, must function using the same physical principles employed by small, natural flying devices such as insects and maple seeds.' He added, 'Our aerodynamic study of maple seeds could help design the first successful powered 'maple' helicopters.'

Countries

Netherlands, United States

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